Comparative Analysis of Filtering for Medical Imaging Modalities: MRI, CT, and X-ray

Introduction

I. Overview

Advancements in medical imaging technologies and the precision of digital filtering techniques are pivotal in transforming diagnostic processes, enhancing the clarity and accuracy of images from modalities such as MRI, CT, and X-ray. As a result, this can significantly influence patient outcomes and the understanding of complex medical conditions. In this paper, I will explore and evaluate the efficacy of various filtering techniques applied to medical imaging modalities—specifically, MRI and CT at first—by manipulating MRI brain data through Python's signal processing libraries to apply ideal filters such as low-pass, high-pass, band-pass, and band-stop. Thus, I will first begin with a detailed analysis of MRI and CT brain images. Initially, I will explore the creation and effects of a Point Spread Function (PSF), simulating its impact on brain images through convolution in the spatial domain and multiplication in the frequency domain, hypothesizing equivalent outcomes and aiming to provide a comparative insight into how different filters influence the visualization and interpretation of medical images. This then sets the stage for a side-by-side comparative study of filtered MRI and CT images, using code and visual aids to highlight differences and similarities in image quality and diagnostic utility. In the second section, the focus will shift to a simulated soft tissue phantom, crafted in Python, featuring varying lesions. Here, I will apply the same filtering techniques under different X-ray fluence levels—low, medium, and high—to observe and quantify how fluence and filtering jointly affect image clarity and the ability to discern pathological features. The final section synthesizes these investigations, providing a comprehensive analysis that connects the practical work conducted to key concepts from our BME271 course. This reflective overview not only underscores the academic insights gained but also discusses the real-world applications of these imaging and filtering techniques in medical diagnostics, emphasizing their relevance and utility in enhancing patient care.

II. Definitions & Datasets

Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) are non-invasive medical imaging techniques. MRI uses strong magnetic fields and radio waves to generate detailed images of the body's internal structures (Berger, 2002). CT uses X-rays to create cross-sectional images of the body (Goldman, 2007). Furthermore, the PSF describes the response of an imaging system to a point source or point object (Boreman, 2001). It characterizes the blur or spread of a point object in an image due to factors such as diffraction, aberrations, and defocus (Smith, 1997). In image processing, 2D convolution is an operation that combines two functions, typically an image and a kernel, to produce a modified image (Gonzalez & Woods, 2002). The 2D Fourier Transform is a mathematical tool that decomposes an image into its sine and cosine components, enabling analysis and manipulation in the frequency domain (Bracewell, 1986). Ideal filters are theoretical filters that perfectly pass or attenuate frequencies in the Fourier domain, as we have learned in class. A low-pass filter passes low frequencies and attenuates high frequencies, while a high-pass filter does the opposite (Oppenheim & Schafer, 2010). A band-pass filter passes frequencies within a specific range and attenuates frequencies outside that range, while a band-stop filter attenuates frequencies within a specific range and passes frequencies outside that range (Lim, 1990). Finally, X-ray fluence is the number of X-ray photons passing through a unit area perpendicular to the beam direction (Bushberg et al., 2012). It is a measure of the intensity of an X-ray beam and is an essential factor in determining the radiation dose delivered to a patient during an X-ray examination (Hendee & Ritenour, 2002).

The MRI data will be sourced from a MATLAB file named "BrainEvan.mat" sent from a friend in a different course who had this file (J Shoemaker, Duke BME), while the CT data will be obtained from an image called "CT.jpeg" from the University of Utah School of Medicine (University of Utah School of Medicine, 2018).

Filtering Analysis of MRI and CT Brain Images

I. 2D and 3D Point Spread Function (PSF)

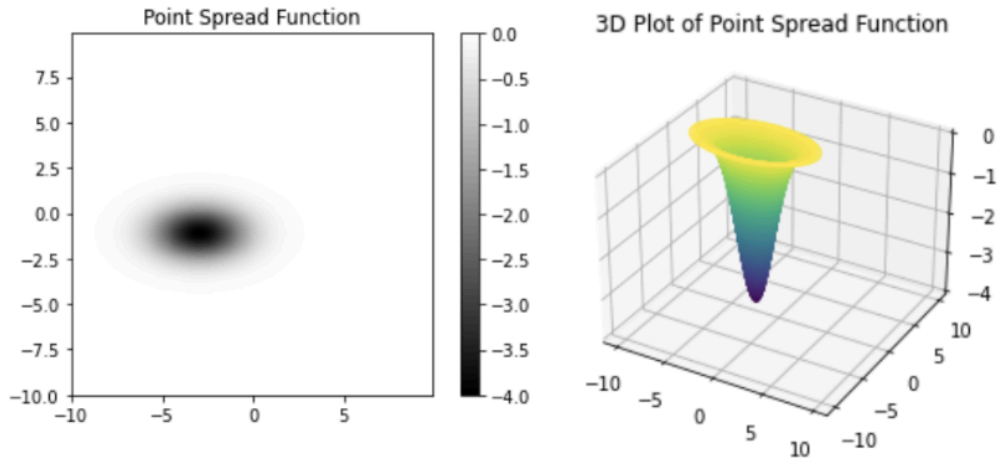


Figure 1. **Visualization of PSF in Imaging.** Left: Two-dimensional representation of a PSF demonstrating intensity distribution. Right: Corresponding three-dimensional visualization of the PSF, illustrating spatial intensity decay from the central peak. Code in Appendix (with explanation).

The 3D representation provides a superior illustration of the system's PSF, offering a clear depiction of its dimensions, including both height and width. This form of visualization showcases the PSF's narrowness, which is indicative of its high resolution. By contrast, the 2D display, while still presenting these details, does so in a manner that can be more challenging to decode. A 3D plot not only reveals the filter type used within the system but also conveys the degree of resolution achievable in multiple orientations. Understanding the width of the graph provides clarity into the function's resolution quality. The analysis of a PSF is closely related to 2D convolution, as the PSF can be used as a convolution kernel to model the imaging system's response and predict the output image when convolved with the input image. This was the reasoning behind exploring PSFs before diving into image processing.

II. 2D Convolution in Spatial Domain vs. Multiplication in Frequency Domain

Figure 2. Equivalence of Spatial and Frequency Domain Operations in Brain Imaging. Left: Brain image after convolution in the spatial domain. Right: Brain image resulting from multiplication in the frequency domain, illustrating that both operations yield the same image due to underlying principles of signals and systems. Code in Appendix (with explanation).

The two methods yield the same result as they essentially perform the same operation in different ways. The first method involves a convolution between the filter and the brain data by taking a Fourier Transform (FT), which then convolves them into the Spatial Domain (SD). The second method, however, is more efficient as it involves separate FTs for the brain data and filters, transforming them individually into the SD. After this, the results are multiplied together in the SD. This difference in approach means that the second method uses multiplication, which is computationally faster than convolution. Despite the different processes, convolution in the Spatial (Image) Domain equates to multiplication in the Frequency Domain, hence the identical results.

III. Ideal Filters: MRI and CT

Figure 3. Filtered CT and MRI Brain Images using Different Filters. Illustrates the effects of Band-Pass (BPF), High-Pass (HPF), Low-Pass (LPF), and Band-Stop (BSF) filters on similar MRI and CT brain images. Code in Appendix (with explanation).

The BPF MRI image shows a balance between preserving some details and reducing noise. The brain structure is visible, but the edges are slightly blurred compared to the original image. The BPF CT image also strikes a balance between detail retention and noise reduction. The brain structure and skull are visible, with some smoothing of the edges. Both BPF images maintain a good balance of detail and noise reduction. The MRI image appears to have slightly more preserved detail compared to the CT image. Moving on, the HPF MRI image emphasizes high-frequency information, resulting in enhanced edges and fine details. The brain structure appears sharper, but the image also has increased noise and a grainy appearance. The HPF CT image also highlights edges and fine details, making the brain structure and skull appear sharper. However, the noise is more pronounced compared to

the MRI image. Both HPF images enhance edges and fine details, but the CT image exhibits more pronounced noise and graininess compared to the MRI image. The LPF MRI image smooths out the original image by suppressing high-frequency information. The brain structure appears blurred, and fine details are lost. The LPF CT image also smooths out the image, resulting in a blurred appearance of the brain structure and skull. Fine details are suppressed, but the noise is reduced. Both LPF images have a blurred appearance due to the suppression of high-frequency information; however, the CT image in LPF is much smoother while much of the MRI image is blurred out which is an interesting conclusion here. The BSF MRI image shows a unique pattern of concentric rings, indicating the suppression of a specific range of frequencies. The brain structure is still visible but with altered contrast and some artifacts that can be seen properly. The BSF CT image also displays concentric rings, suggesting the attenuation of a specific frequency range. The brain structure and skull are visible but with more blurriness and visuals are not the best. Both BSF images exhibit similar concentric ring patterns, indicating the suppression of a specific frequency range. The MRI and CT images show comparable levels of altered contrast and artifacts. In summary, the BPF images maintain a balance between detail and noise reduction, with the CT image having slightly more preserved detail. The HPF images enhance edges and fine details but introduce more noise, particularly in the CT image. The low-frequency components of the brain's image fade out in both the BPF and HPF, which makes sense as they filter out lower frequencies. The LPF images have a blurred appearance due to the suppression of high frequencies, with more smoothing in the CT, even though the MRI image resembles the original image closely. The BSF images show concentric ring patterns, indicating the attenuation of a specific frequency range, with comparable artifacts in both modalities. The application of ideal filters to MRI and CT images of the brain highlights how different frequency components contribute to the visualization of specific brain structures. Low-pass filters, which attenuate high frequencies, result in the suppression of fine details and edges, causing the brain structures to appear blurred. This blurring effect is due to the removal of high-frequency information that corresponds to sharp transitions and intricate patterns within the brain. On the other hand, high-pass filters, which attenuate low frequencies, emphasize edges and fine details by preserving the high-frequency components. This emphasis on high frequencies leads to enhanced visibility of brain structures with sharp boundaries, such as the skull and ventricles, but also amplifies noise and creates a grainy appearance. The selective filtering of specific frequency ranges through band-pass and band-stop filters further demonstrates the importance of different frequency components in representing various brain structures and how their manipulation can alter the visual representation of the brain in MRI and CT images.

IV. Extension with Convolution Matrix Application

Figure 4. Convolution Analysis of CT Brain Imaging Using Different Kernel Sizes. Brain image obtained through convolution in the spatial domain with varying kernel sizes, showing the effect of convolution with an increasing kernel matrix size on image clarity and feature visibility. Code in Appendix (with explanation).

This figure demonstrates the progressive blurring of a CT scan image through the application of different-sized filters, simulating the effect of kernel size on image clarity. Each image showcases the results of applying a two-dimensional convolution with square matrices of varying sizes, illustrating how increasing the filter size (NxN) leads to a noticeable reduction in detail and sharpness due to enhanced blurring. In medical imaging applications, optimizing image clarity is crucial for accurate diagnosis, particularly when noise reduction or specific feature enhancement is needed.

X-ray Fluence and Filtering

In this section, I will dive into the interplay between X-ray fluence and filtering techniques to explore their effects on diagnostic imaging to continue my investigation with filtering and medical imaging modalities. Using Python, I simulated a soft tissue phantom to emulate the characteristics of human tissues, and artificial lesions were introduced to represent tumors, creating a controlled environment for studying image quality. By applying various ideal filters and manipulating X-ray fluence levels, this section seeks to identify the optimal combination of fluence and filtering that enhances lesion detectability within the phantom. The findings will provide valuable insights into how these parameters impact the clarity and diagnostic accuracy of X-ray imaging in identifying subtle pathological features.

I. Phantom and Fluence

Figure 5. Effects of X-ray Fluence on Lesion Detection in Simulated Soft Tissue Phantom. The leftmost image shows the original phantom with embedded lesions. The subsequent images illustrate the impact of different X-ray fluence levels on the visibility of these lesions, demonstrating that higher fluence enhances lesion visibility while lower fluence results in increased noise, thereby obscuring the lesions. Code in Appendix (with explanation).

X-ray fluence, a measure of the X-ray photon density per unit area, directly influences the quality and clarity of imaging. When the fluence is set to 5, the limited number of photons penetrating the tissue phantom only provides enough contrast to make the four largest lesions visible, while smaller lesions are obscured by noise due to insufficient photon count. Reducing the fluence to 2 significantly lowers the number of photons reaching the detector, exacerbating noise and causing a dramatic reduction in image clarity. This results in only the two largest lesions being distinguishable, as the signal from smaller lesions is completely overshadowed by the increased noise. Conversely, increasing the fluence to 10 enhances the signal-to-noise ratio, thereby improving image clarity and allowing for the visualization of more lesions. With more photons available to interact with the tissue phantom, the resulting image provides enough contrast to reveal six of the nine lesions, including smaller ones that were previously obscured. This shows the crucial impact of X-ray fluence on diagnostic accuracy, as adjusting fluence levels can greatly affect the detectability of lesions, especially smaller and less dense ones.

II. Filtering

Figure 6. Filtered Phantom Images with Different Fluence Levels and Filtering Techniques. Demonstrates the impact of different fluence levels and filtering techniques on soft tissue phantom images. The first row shows the original phantom and its processed versions after applying various filters. The second and third rows display the effects of different fluence levels, ranging from 2 to 10 respectively, on the visibility of the lesions in the phantom, with notable changes observed in clarity and lesion detectability across different filter types.

Analyzing the images of the simulated soft tissue phantom at various fluence levels and with different filters offers insight into the optimal conditions for detecting lesions or tumors. The lesions themselves can be observed best with the HPF, as the outlining of the tumors make them contrast well with the background. Additionally, the LPF and BSF produce similar images here which is an interesting observation. Starting with the original phantom, lesions are visually discernible under moderate fluence levels. At lower fluence levels (Fluence = 2), noise becomes prominent, obscuring the lesions. However, increasing the fluence to 10 makes six of the nine circles visible, but further noise is evident. Comparing the images under various filters, LPF blurs out finer details due to the elimination of high-frequency components, causing loss of image sharpness, which makes it challenging to detect smaller lesions, especially at low fluence levels. Conversely, the BPF and BSF retain more features by selectively filtering specific frequency bands, helping to maintain a balance between noise reduction and image clarity. Of these, the BSF at fluence 10 provides the most distinguishable view of the lesions, preserving their edges and clarity by blocking specific bands that obscure detail. The BPF, while effective in high-contrast conditions, introduces noise at low fluence levels due to less effective filtering and we are unable to observe any lesions whatsoever. The combination of fluence level and filter type plays a significant role in optimizing lesion detectability. For instance, a high fluence of 10 with a BSF maintains the greatest clarity, highlighting more lesions distinctly compared to other combinations, such as low fluence and HPF/LPF/BPF, which obscure the image due to excessive noise.

Integrating Practical Insights and Theoretical Concepts in Medical Imaging

Throughout this project, I have explored various practical aspects of medical imaging, specifically focusing on MRI, CT, and X-ray modalities. In doing so, I have gained valuable insights into the application of key concepts from our BME271 course, which has deepened my understanding of the theoretical foundations underpinning these imaging techniques.

One of the fundamental concepts we explored in the course was the representation of systems using convolution. The practical analysis of the PSF in imaging systems highlighted the importance of this concept. By visualizing the 2D and 3D representations of the PSF, I gained a deeper appreciation for how the system's response to a point source or object is characterized by the spread or blur in the resulting image. This direct connection to the concept of convolution, where the PSF acts as a convolution kernel, allowed me to model the imaging system's response and predict the output image when convolved with the input image. Moreover, the equivalence between convolution in the spatial domain and multiplication in the frequency domain, as demonstrated by the convolutional analysis of brain images, reinforced the principles of signals and systems that we studied in the course. The application of ideal filters to MRI and CT brain images provided a concrete example of how different frequency components contribute to the visualization of specific brain structures. By observing the effects of low-pass, high-pass, band-pass, and band-stop filters, I gained a deeper understanding of how these filters manipulate the frequency content of the images, leading to varying degrees of detail preservation, noise reduction, and edge enhancement. Moreover, I realized that the filter to use depends on what exactly one wants to look at, especially in the case of these brain images. This practical experience aligned with the theoretical concepts of Fourier analysis, where signals can be decomposed into their frequency components and selectively filtered based on the desired frequency range. The ability to visualize the impact of these filters on brain images solidified my comprehension of how different frequency components represent different aspects of the image, such as fine details, edges, and noise.

Furthermore, the analysis of convolution with varying kernel sizes in CT brain imaging emphasized the importance of optimizing image clarity for accurate diagnosis. This practical exploration directly relates to the concepts of incrementally linear systems and their convolution representation, where the system's output is determined by the convolution of the input signal with the system's impulse response. By observing the progressive blurring of the CT scan image with increasing kernel sizes, I gained a deeper appreciation for how the characteristics of the convolution kernel can significantly influence the quality and clarity of the resulting image. Moving on, the investigation of X-ray fluence and filtering techniques in the context of a simulated soft tissue phantom provided valuable insights into the interplay between these parameters and their impact on lesion detectability. By manipulating fluence levels and applying different filters, I could observe the effects on image clarity, noise reduction, and the visibility of pathological features. This practical experience reinforced the theoretical concepts of signal-to-noise ratio, frequency domain analysis, and the importance of optimizing imaging conditions for diagnostic accuracy. The ability to visualize the trade-offs between fluence levels, noise, and lesion visibility solidified my understanding of how these factors influence the quality and diagnostic value of medical imaging.

Throughout this project, I have not only gained practical experience in applying various imaging and filtering techniques but have also reinforced and deepened my understanding of the theoretical concepts covered in our BME271 course. The connections made between the practical work and the underlying principles of signals and systems, Fourier analysis, convolution representations, and system characteristics have provided a holistic perspective on medical imaging. This integration of theoretical knowledge and practical application has enriched my comprehension of the subject matter we learned and has highlighted the relevance and utility of these concepts in enhancing patient care through accurate and precise medical diagnostics.

Appendix

```python
# Plotting the 2D Point Spread Function
import numpy as np
import matplotlib.pyplot as plt

# Setting dx and dy as step sizes
dx = 0.1
dy = 0.1

# Creating 1D arrays using arange
x_range = np.arange(-10, 10, dx)
y_range = np.arange(-10, 10, dy)

# Creating meshgrid matrix for plotting image
x, y = np.meshgrid(x_range, y_range)

# Plotting impulse response function using imshow and the equation given above
h1 = -4 * np.exp(-((x+3)**2)/6-((y-1)**2)/2)
fig, ax = plt.subplots()

my_plot = plt.imshow(h1,
                     cmap = "gist_gray",
                     vmin = h1.min(),
                     vmax = h1.max(),
                     extent =[x.min(),
                              x.max(),
                              y.min(),
                              y.max()])
fig.colorbar(my_plot, ax = ax)
ax.set_title('Point Spread Function')
plt.show()

# Plotting the 3D Point Spread Function

fig2 = plt.figure()
ax = plt.axes(projection='3d')
ax.contour3D(x, y, h1, 900)
ax.set_title('3D Plot of Point Spread Function')
```

Code on the left defines a 2D Gaussian kernel to simulate the Point Spread Function (PSF), and uses ‘imshow’ and ‘contour3D’ to visualize the PSF in 2D and 3D plots. Code on the right compares convolution of the brain image with a PSF in the spatial domain to multiplication in the frequency domain, demonstrating that both approaches yield equivalent results by visualizing the processed images. The code achieves this by performing a Fast Fourier Transform (FFT) on both the image and the PSF, multiplying them in the frequency domain, and then applying an inverse FFT to obtain the spatial domain result.

```python
# Filter Creation - Band Pass Filter

x_range = np.linspace(-5, 5, len(brain))
y_range = np.linspace(-5, 5, len(brain))

Rin = 0.5
Rout = 1

#Meshgrid
x, y = np.meshgrid(x_range, y_range)

# Concentric
array1 = (((x**2 +y**2)> Rin ** 2)&((x**2 + y**2)< Rout ** 2)) * 1

# Fourier Transform (FT)
filt1 = np.abs(np.fft.fftshift(np.fft.ifft2(array1)))
FFT_brain = np.fft.fftshift(np.fft.fft2(brain))
FFT_tot = array1 * FFT_brain

# FT Inversion
inv_shift = np.abs(np.fft.fftshift(np.fft.fftshift((np.fft.ifft2(FFT_tot)))))

# Overall Title

plt.figure(figsize=(17, 5))
plt.suptitle('Band Pass Filter')


plt.subplot(131)
plt.imshow(inv_shift, cmap = "gist_gray")
plt.title('BPF Image')
plt.axis([0,250, 250, 0])

plt.subplot(132)
plt.imshow(array1, cmap = "gist_gray")
plt.title("BPF (Frequency)")
plt.axis([0,250, 250, 0])

plt.subplot(133)
plt.imshow(filt1, cmap = "gist_gray")
plt.title("BPF (Image)")
plt.axis([100,150, 100, 150])

plt.show()
```

```python
from PIL import Image
from scipy.signal import convolve2d

CT = np.array(Image.open('CT.jpeg').convert('L'), dtype=np.float64)

def convolve_NxN(data, N):
    kernel = np.ones((N, N)) / N**2
    CT_blur = convolve2d(data, kernel, mode='same', boundary='symm')

    plt.figure()
    plt.imshow(CT_blur, cmap='gray')
    plt.title(f'Convolution with {N}x{N} matrix')
    plt.axis('off')
    plt.show()

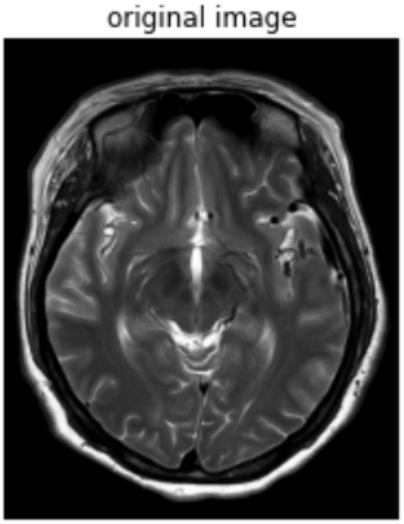
convolve_NxN(CT, 100)
```

Code on the left generates a band-pass filter, applies it to the MRI brain image in the frequency domain through Fourier transforms, and visualizes the filtered image alongside its frequency response. This code can be generalized to the other code snippets for the remaining filters. Code on the right applies a convolution matrix to a CT image,

using a kernel to perform convolution and visualize the resulting image blurring as the kernel matrix size increases (Adapted from Dr. Micheal Rizk's Convolution and Images Discussion file).

Finally, the code snippet above generates simulated soft tissue phantoms and adjusts their X-ray fluence levels importantly, enabling analysis of different fluence settings on lesion visibility within the images. It achieves this by setting different photon densities ('Lambda' values) to simulate fluence, which adds varying levels of Poisson noise to the images. Poisson noise, in this context, simulates the inherent statistical variations in X-ray photon counts due to limited photon flux, reflecting realistic imaging noise observed in low-fluence medical imaging conditions. The intensities represent the contrast between the lesions and surrounding tissue, with different intensity values assigned to the lesions and background to simulate the visibility of tumors or abnormalities in the images.

Original CT Image and MRI Dataset Explanation



The MATLAB file, 'brainEvan.mat', contains a 256x256 matrix that represents preprocessed brain MRI data, which include a raw MRI image/ processed image arrays. This matrix is used as input data for subsequent image processing tasks such as filtering, convolution, and analysis, providing a structured and consistent data set for evaluating different image processing techniques as observed in this report.

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